

KANAD S.H.I.E.L.D. 2026 · AHMEDABAD CITY POLICE · I-HUB GUJARAT

Paritran

FROM COMPLAINT TO CONVICTION

1.6%

the cybercrime conviction rate India is stuck at. Paritran builds the court-ready case file that detection tools never produce.

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PS-69EEFE4F8CD1C · Category 2 · Problem Statement 10 (Open-Ended Smart Policing)

THE PROBLEM

Detection is solved. Conviction is not.

₹22,845cr

LOST TO CYBER FRAUD, 2024

2.43%

NCRP REPORTS → FIRS

18%

CHARGESHEETING RATE

1.6%

CONVICTION RATE

**India's cyber-justice gap
18.1%**

2.43%

NCRP → FIR



Chargesheet

1.6%

Conviction

Three things turn volume into almost no justice

● The money is gone before anyone acts

Fraud moves across mules in minutes; banks take ~6 days. ₹8,189 cr saved of ₹20,000 cr.

● The case is never built to court standard

Hundreds of small complaints belong to a few syndicates, never linked; evidence lacks a clean custody chain.

● The legal bar just went up

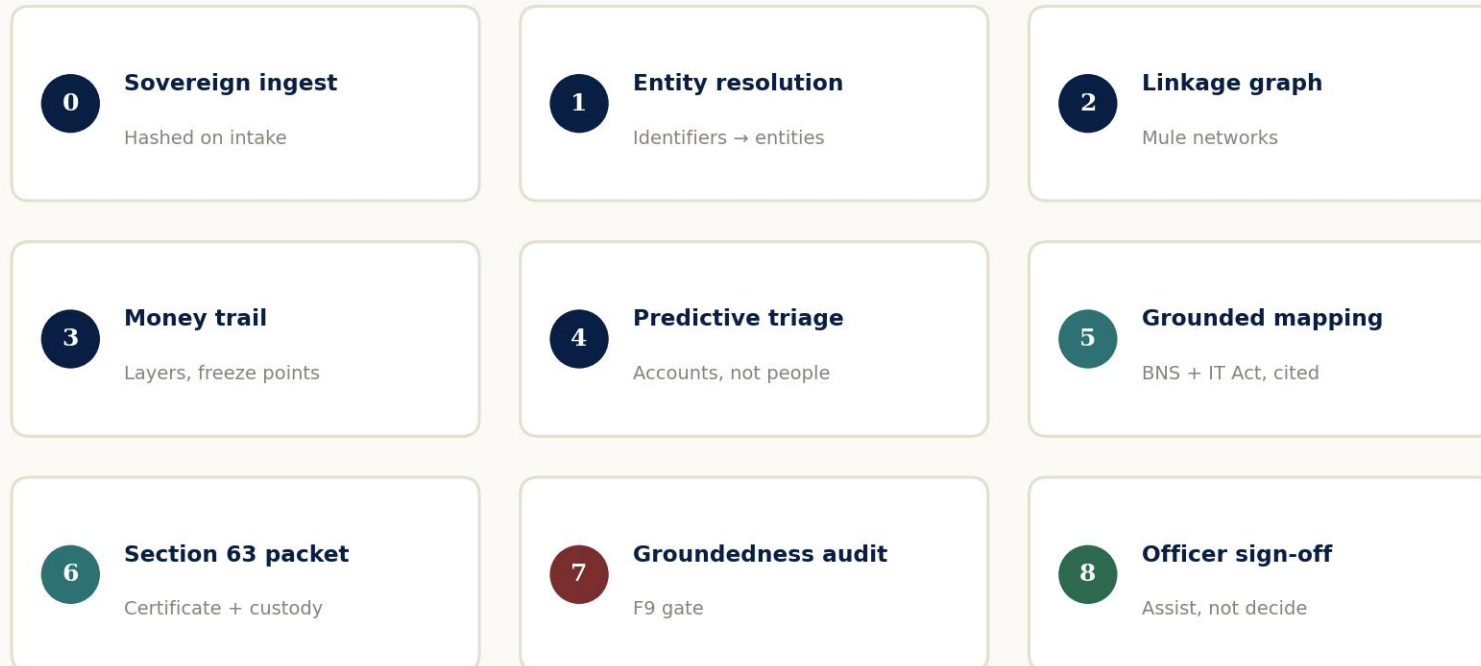
Since 1 July 2024, BNS, BNSS and BSA replaced IPC/CrPC/Evidence Act. Section 63 now needs dual certification.

Gujarat: about ₹678 cr from ~72,000 victims (Jan-Sep 2025). Ahmedabad logged the most cyber-fraud calls in the state, 28,382.

PROPOSED SOLUTION

The case file that holds up in court

Paritran runs inside the branch. It links complaints into mule networks, rebuilds the money trail with a tamper-evident custody chain, drafts the Section 63 certificate, and maps every offence to the right law, quoted word for word. The officer reviews and signs. The engine never decides.



Why it wins

- Court-ready, not just detected**

We own the packet that ends in a conviction.

- Deterministic core, fenced model**

Same input, same output; the model only drafts, gated.

- On-premise, zero egress**

Victim data never leaves the branch. Works in Gujarati.

- Accounts, not people**

Scores money and infrastructure; no profiling.

SYSTEM ARCHITECTURE

One deterministic core, a fenced language layer



Ingest layer

Text + voice, multilingual · SHA-256 hash on intake · zero egress



Intelligence core (deterministic)

Entity resolution · linkage graph · money-trail reachability · triage



Grounded language layer (fenced)

BM25 + InLegalBERT + rules agree · model only drafts cited evidence



Assurance layer

F9 groundedness gate · SHA-256 hash-chained custody · officer sign-off



Deployment

On-premise / gov-cloud · Docker offline bundle · API-first CCTNS + ICJS

Holds up under scrutiny

- Offline-first, runs air-gapped in a station
- Deterministic core: reproducible at a fixed seed
- PII stays inside the branch boundary
- Model down → retrieval + rules + human review
- Hash-chain anchored out of band
- Concurrency-safe append-only audit ledger

TECHNOLOGY STACK

Built to run inside a police station

● Frontend

- React + TypeScript + Vite
- react-force-graph, Motion
- cmdk palette, react-countup
- TanStack Query, SSE live

● Backend + data

- Python 3.11, FastAPI
- PostgreSQL + RLS audit
- NetworkX graphs
- Plain-SQL migrations

● AI / ML

- InLegalBERT + BM25 retrieval
- Rule + retrieval agreement
- Gemma via Ollama (local)
- spaCy NER, Tesseract OCR

● Security + ops

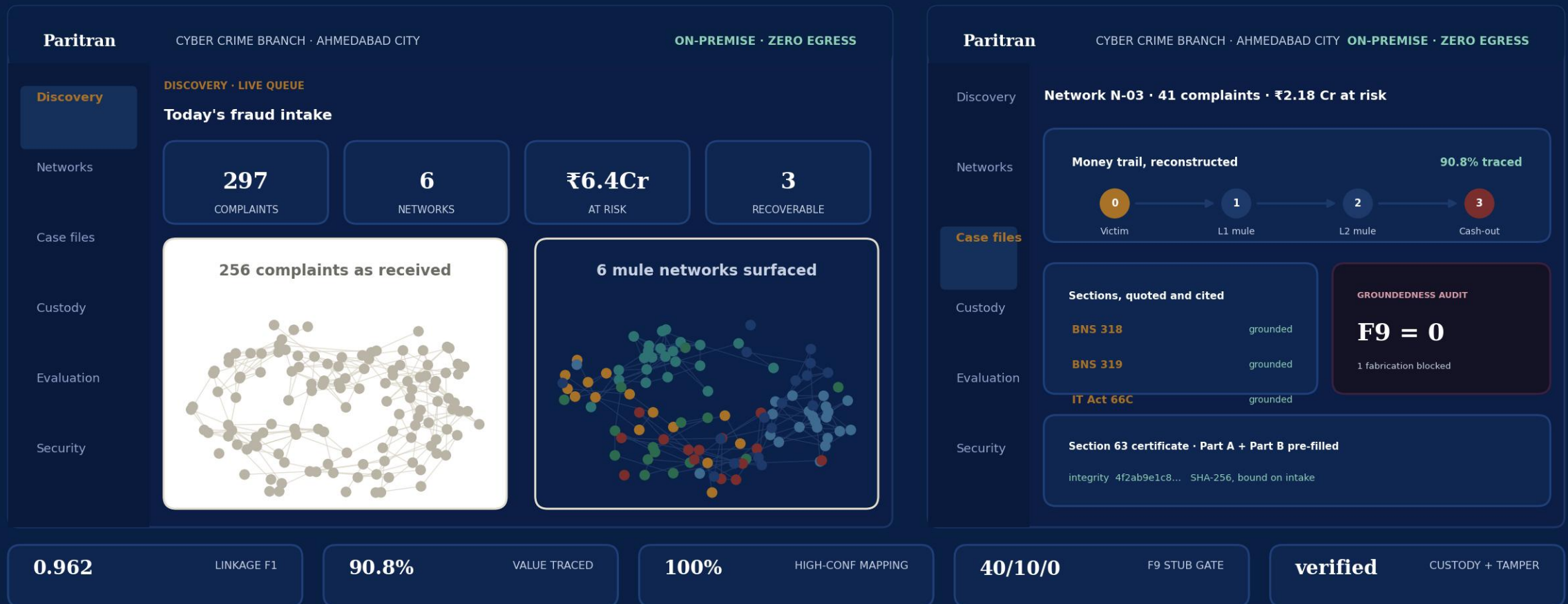
- 7 scanners in CI
- OWASP ASVS 5.0
- Prometheus + Grafana
- Docker offline bundle

On-premise on purpose. Complaint and victim data never leave the branch, under DPDP and data-sovereignty norms. We treat that as a requirement and a strength.

DEMONSTRATION

Watch it work

Product views from the working prototype, filled with live seed-42 output. The chrome is designed; the numbers, graphs and hashes are real.



INNOVATION

The two rows nobody else ticks

Capability	I4C / Pratibimb	RBI MuleHunter	StaQu CrimeGPT	Legal AI	Paritran
Fraud & mule detection	Yes	Yes	Search	No	Consumes
Court-ready Section 63 packet	No	No	No	No	YES
Tamper-evident custody	No	No	No	No	YES
Grounded mapping, no fabrication	N/A	N/A	Generative	Partial	F9 gated
Runs on-premise, zero egress	Gov cloud	Bank	Claimed	Cloud	Yes

Precision via abstention

100%

accuracy when three independent paths (rule, BM25, InLegalBERT + model) agree, on 8 of 8 and 15 of 15 golden cases.

Everything else routes to a human.

The moat: the Section 63 admissible packet and the tamper-evident custody chain. Everyone optimises detection. Paritran owns the part that ends in a conviction.

Why not just use ChatGPT? A police station needs offline, sovereign, legally-grounded, auditable, local-language output that supports admissibility. A public chatbot gives none of these.

IMPACT

Hours to seconds, money kept in reach



In parallel: golden-hour triage → higher freeze rate → more money recovered.



Public value: faster, lawful action for victims, and officer hours returned to investigation instead of paperwork.

SCALABILITY & FUTURE SCOPE

A path, and a companion

1 Phase 1 BUILT

Demo slice

Ingest, linkage, money trail, Section 63, audit gate. Built and measured.

2 Phase 2 NEXT

Branch pilot

Standalone on-premise, Gujarati drafting, officer UX. 90 days.

3 Phase 3 DESIGNED

Integrated

API-first with CCTNS, ICJS, CFCFRMS as interfaces open.

One core, two faces

The same grounded-mapping and audit engine also answers the crime-documentation statement. Standalone on-premise on day one.

Sustainable and procurable

Deployable via GeM and empanelment; hosting and maintenance costed for the branch. Winners get a deployment pathway with the Police.

SUPPORTING DOCUMENTS & RESOURCES

Verify everything in ten seconds

Fixed seed. Reproducible. Synthetic data only, zero real PII. Scan any code and run it yourself.



Source code / repository



Live website



Demo video



Solution document (PDF)



Research white paper



Reproduce results



Dataset datasheet



Contact Team Paritran

Ready to pilot with the Cyber Crime Branch

Paritran. From complaint to conviction.

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Thank you. Questions welcome.

Measured, reproducible, honest

0.962

LINKAGE F1

P 0.957 · R 0.966

90.8%

VALUE TRACED

graph reachability

100%

MAPPING, HIGH-CONFIDENCE

8/8 · 15/15, rest routed

0

F9 LEAKED

40/10/0 stub · live gated

0.74/1.0

NER P/R

seed 42

verified

CUSTODY + TAMPER

SHA-256 chain

Reproduce it on any seed you name

191 tests green · 34-second offline demo · seed 42 regenerates every number.
A judge types any seed in the Evaluation screen and every value moves; no canned number survives.
pip install networkx==3.4.2 && python3 src/paritran_prototype.py reproduces results.json standalone.

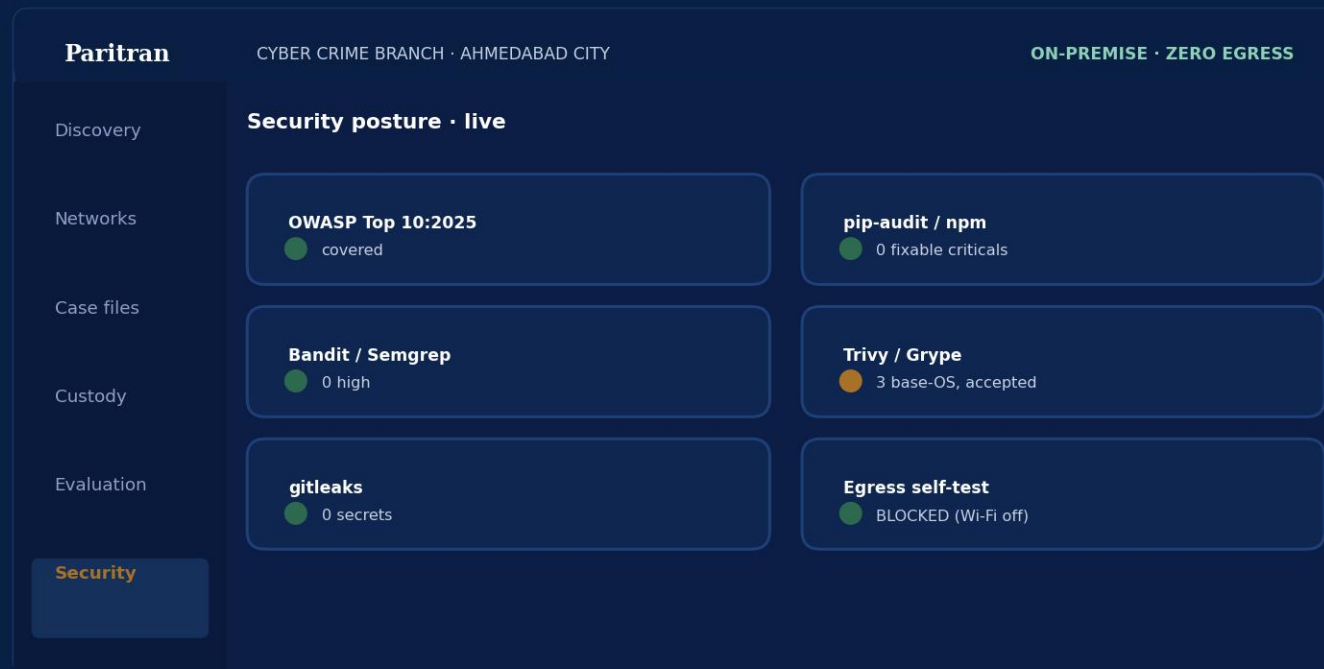
We corrected our own numbers

During prep we audited our own submitted appendix, found three numbers produced by placeholder code, replaced them with the real methods, re-measured, and published both columns in a public changelog. Our numbers went from claimed to reproducible on any seed.

	WHAT WE SUBMITTED		WHAT WE NOW RUN
Money trail	87.3% random draw	→	90.8% graph reachability
Section mapping	90.5% keyword ladder	→	52.4% floor · 100% high-confidence
F9 groundedness	0/50 tautology	→	gated over live model, 0 leaked
Linkage F1	0.936	→	0.962 re-measured

Honesty is the deliverable. The one genuine vulnerability, owned in the open, becomes the strongest trust signal in the room.

Security you can see, not just claims



Observability + privacy

- Prometheus + Grafana, 8-panel dashboard
- OpenTelemetry traced, custom metrics live
- Live system-status widget over SSE
- DPDP-aligned, on-premise, zero egress
- Encryption at rest (host FDE), TLS for pilot
- Hash-chain head anchored out of band

One button, five beats, offline

- 1 Intake** 297 complaints stream in; counters climb, ₹ at risk rises.
- 2 Collapse** Force graph resolves 256 into 6 mule networks; officer rejects one link (hashed).
- 3 Money trail** Value flows victim → cash-out; traced % climbs to 90.8.
- 4 Packet + F9** Section 63 assembles with verbatim quotes; a planted fabrication is blocked live.
- 5 Custody** Hash chain renders; tamper test breaks a scratch chain, real chain stays verified.

Runs with Wi-Fi off, the zero-egress proof, live. Fallbacks: captioned video and a stub-degrade banner, all numbers stay live.
Reproduce standalone: `python3 src/paritrان_prototype.py`

Evidence integrity, and reproduce it yourself

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CYBER CRIME BRANCH · AHMEDABAD CITY

ON-PREMISE · ZERO EGRESS

Discovery

Chain of custody · verified across 12 records

Networks

Case files

Custody

Evaluation

Security

#	ARTEFACT	SHA-256	PREV	LINK
00	evidence_00	ed0f4c...	bf22bd...	verified
01	evidence_01	88c5d7...	46eea3...	verified
02	evidence_02	5f4e64...	034b74...	verified
03	evidence_03	ad3900...	ed6ce8...	verified
04	evidence_04	296089...	ab0531...	verified
05	evidence_05	14f04a...	c20edb...	broken
06	evidence_06	56bd75...	e76f49...	verified

Tamper test: record 06 altered, chain breaks at 06 and is flagged. Real chain stays verified.

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CYBER CRIME BRANCH · AHMEDABAD CITY

ON-PREMISE · ZERO EGRESS

Discovery

Evaluation · reproduce & judge-seed

Networks

Case files

Custody

Evaluation

Security

Linkage F1

Value traced

Mapping, high-confidence

F9 leaked

NER precision / recall

Reproduce (seed 42)

Judge's seed: __

The custody chain is tamper-evident and the whole pipeline reproduces at a fixed seed. A judge can break a record and watch it flag, or type any seed and watch every number move.